

An Autonomous Completion Marker and Dressed-Bath Audit for a Conditional Latch

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Abstract—We replace an initially enabled latch gate with a static terminal-clock coupling that flips a fresh completion flag. A native ququart and two qubits form a two-pair interaction graph. Exact invariant-sector identities establish completion-marked capture and exclusion of a designated blocked sector. A directed interval calculation retains the original local-bath readout certificate. Separately, an Ohmic exponential-cutoff spectrum defines a dressed thermal generator. Its fast-rate replay fails a declared secular-separation diagnostic, while a weak-rate replay passes that diagnostic and requires a much later sampled deadline. These conditional results do not certify frontend work-word compilation, a complete qubit two-local processor or microscopic bath accuracy.

Index Terms—Completion markers, local interactions, thermal reservoirs, dressed transitions, MQM.

I. TYPED INTERFACE AND FROZEN PROTOCOL

The preceding explicit-reservoir latch presumes an eligible launch with its m - e coupling enabled. Here let C be a native ququart with states p, m, e, d , F a fresh completion qubit and R a damped auxiliary qubit. Launch is $|p, 0, 0\rangle$ and retained target is $|d, 1, 0\rangle$. Stored data and arbitrary reference spectators are retained; every port term acts as identity on them. The bath is traced.

The frontend contract assigns p to the terminal address after its work word. This paper generates the terminal marker from that address, rather than assuming an initially true flag. It does not prove that a compiled native quantum work word reaches p , nor implement an occupancy sensor or capacity-to-completion trigger. Those are separate admission gates.

Before evaluation, freeze $h = g = 1$, $\lambda^2 = 4\sqrt{2}$, local damping $\kappa = 8$, local reverse rate zero, and original window $|t - 16| \leq 1/1024$. For a separate dressed-bath audit fix $\Omega = 64$, exponential cutoff $\omega_c = 640$, $\beta\Omega = 20$, reference downward rates 8 and $1/1024$, and times 16, 256, 65536, 262144, 1048576. All frequency units are relative to g ; none are calibrated hardware units.

II. STATIC MARKER AND EXACT SECTOR SAFETY

With $P_1 = |1\rangle\langle 1|_F$, use

$$\begin{aligned} H = & h(|m\rangle\langle p| + \text{h.c.}) \otimes X_F \otimes I_R \\ & + g(|e\rangle\langle m| + \text{h.c.}) \otimes P_1 \otimes I_R \\ & + \lambda(|d\rangle\langle e| \otimes I_F \otimes \sigma_R^+ + \text{h.c.}). \end{aligned} \quad (1)$$

The local jumps are $\sqrt{\kappa}I_{CF} \otimes \sigma_R^-$ and, when included, $\sqrt{\kappa}rI_{CF} \otimes \sigma_R^+$. No pulse scheduling is required by this cell

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Hamiltonian. Both interaction pairs, C - F and C - R , are native two-subsystem couplings.

The admitted projector is

$$P_{\text{adm}} = [P_p \otimes P_0 + (P_m + P_e + P_d) \otimes P_1] \otimes I_R. \quad (2)$$

It has rank eight, commutes with H and both bath ladder operators, and contains the launch. Capture inside its range therefore always carries $F = 1$. A separate rank-four blocked projector,

$$P_{\text{blk}} = [P_p \otimes P_1 + P_m \otimes P_0] \otimes I_R, \quad (3)$$

is also invariant and excludes d . Thus the designated blocked sector cannot capture, even with reverse jumps. This is a typed sector statement, not protection against arbitrary erroneous preparations such as $|e, 0, 0\rangle$ outside these launch contracts.

Nineteen exact checks establish these identities, the rank-eight embedding, charge conservation, spectra and an encoded Pauli obstruction. Fifty-two binary64 checks audit the bath decomposition, positivity, normalization, detailed-balance ratios, stationary Gibbs substitution and resource norms. These counts are reproducibility receipts, not independent review.

The native Hilbert-space dimension is $4 \cdot 2 \cdot 2 = 16$. Its exact admitted reduction has dimension eight; this does not remove physical subsystems. In the binary encoding $p = 00, m = 01, e = 10, d = 11$, the marker contains $hZ_{C1}X_{C2}X_F/2$. Therefore this displayed four-qubit encoding has a nonzero weight-three term. Native pair locality does not establish a qubit two-local compiler.

III. FULL SPECTRUM AND LOCAL-BATH READOUT

Writing x for the spectral variable, the unrestricted interaction characteristic polynomial is

$$\begin{aligned} \chi_H(x) = & x^4(x^2 - h^2)^2(x^2 - \lambda^2)(x^2 - g^2 - h^2) \\ & \cdot [x^4 - (g^2 + h^2 + \lambda^2)x^2 + h^2\lambda^2]. \end{aligned} \quad (4)$$

At the selected parameters, the quartic discriminant is exactly $(2 + 4\sqrt{2})^2 - 16\sqrt{2} = 36$. The largest squared eigenvalue is $4 + 2\sqrt{2}$, giving full centered interaction norm $\sqrt{4 + 2\sqrt{2}} \approx 2.613126$. This includes blocked and other sectors; it is not merely a low-energy projection budget.

For the cold local bath, the live path is $p00, m10, e10, d11$. Its real no-jump generator is

$$A = \begin{pmatrix} 0 & -h & 0 & 0 \\ h & 0 & -g & 0 \\ 0 & g & 0 & -\lambda \\ 0 & 0 & \lambda & -\kappa/2 \end{pmatrix}. \quad (5)$$

The amplitudes differ by successive powers of $-i$. Every downward jump parks the system in $d10$. Thus the retained fixed-launch preparation-channel half-diamond error equals the squared norm of the surviving four-component vector. This equality retains the flag, auxiliary and arbitrary spectator data, while tracing the original bath.

An 80-decimal-digit directed interval calculation, using Taylor order 32, scaling by 2^{14} , an analytic infinity-norm remainder and interval squaring, verifies throughout the original window

$$q(t) < \frac{5}{1048576} < \frac{1}{65536}. \quad (6)$$

The interval upper endpoint is approximately $4.636 \cdot 10^{-6}$; the center-time numerical error is approximately $3.924 \cdot 10^{-6}$.

The CPTP Duhamel reverse-jump debit is at most $\kappa r t$. Consequently the sufficient range $r \leq 2^{-24}$ gives

$$\begin{aligned} \delta(t) &\leq \frac{5}{2^{20}} + 8 \cdot 2^{-24}(16 + 1/1024) \\ &= \frac{26625}{2147483648} < \frac{1}{65536}. \end{aligned} \quad (7)$$

Here $26625 = 3 \cdot 5^3 \cdot 71$, while the common target numerator is 32768. This is exact error-budget arithmetic, not a new geometric identification. The rational ceiling and sufficient reverse range are selected from derived bounds after exploration and checked separately; no independent prediction is asserted. This certificate applies to the local generator, not automatically to the dressed bath below.

IV. CONCRETE DRESSED THERMAL-BATH MODEL

To test the local approximation, declare the resonant free Hamiltonian

$$H_0 = \Omega[(P_p + P_m + P_e) \otimes I_{FR} + I_{CF} \otimes N_R]. \quad (8)$$

It commutes with the interaction. Couple $A_B = I_{CF} \otimes X_R$ to a bosonic thermal bath, with rate-normalized spectral density

$$J(\nu) = \alpha \nu e^{-\nu/\omega_c}, \quad \gamma_{\downarrow}(\nu) = \frac{J(\nu)}{1 - e^{-\beta\nu}}, \quad \gamma_{\uparrow} = e^{-\beta\nu} \gamma_{\downarrow}. \quad (9)$$

The constant absorbs the spectral-density convention and is chosen so $\gamma_{\downarrow}(\Omega) = \kappa_{\text{ref}}$. This is a declared continuum model, not a measured material spectrum. Lamb shifts are uncalculated and assumed compensated; they remain a separate error debit.

Use projectors of $H_0 + H$ to form frequency-resolved lowering operators $A(\nu) = \sum_{E' - E = \nu} \Pi_E A_B \Pi_{E'}$. Equal Bohr frequencies are grouped into one operator rather than split into independently resolving bath channels. This constructs a fully secular, dressed, Davies-type generator conditional on Born–Markov and secular assumptions. The distinction between local and global equations is regime dependent [1], [2]; neither is automatically an exact microscopic reference.

The admitted sector has twelve active positive frequencies, ranging approximately from 59.972661 to 68.027339. Each has its own reverse factor $e^{-\beta\nu}$; a single rate ratio at Ω cannot simply be substituted for them. The generator preserves the admitted sector, and its Gibbs state is the Gibbs state restricted to that sector. It is not an ergodic Gibbs state over all sixteen

TABLE I
DRESSED-BATH REPLAY; CONDITIONAL MODEL, NO HARDWARE UNITS

κ_{ref}	Separation ratio	$\delta(16)$	$\delta(2^{18})$
8	15.1650	$3.6891 \cdot 10^{-6}$	$9.9005 \cdot 10^{-9}$
1/1024	0.0018512	0.998340	$9.8650 \cdot 10^{-9}$

physical states. The numerically verified target error of this restricted Gibbs state is approximately $9.865 \cdot 10^{-9}$.

V. SEPARATION DIAGNOSTIC AND SAMPLED LATENCY

The frozen heuristic gate requires the largest individual resolved-jump linewidth divided by the minimum distinct positive-frequency separation to be below 1/10. This is only a scale-separation diagnostic; it does not bound Born–Markov errors, bath memory, summed linewidths or secular approximation error. The fast-rate secular replay numerically passes the target at 16, but fails the separation gate. Its output cannot justify a physical fast-bath certificate or falsify a local singular-coupling approximation. The weak-rate case passes the diagnostic, fails the target at 16, 256, 2^{16} , and first passes among the frozen samples at 2^{18} . This is a sampled deadline, not a minimum latency or a certified time window.

VI. PROMOTION BOUNDARY AND NEXT GATE

The maximum laboratory energy is $2\Omega + \sqrt{2}$ and the minimum is zero, so the full laboratory centered norm is $64 + \sqrt{2}/2 \approx 64.707107$. Under an equivalent unit-norm rescaling, Hamiltonian and rates must both scale, and elapsed time increases by this factor. A small interaction-picture norm must not hide laboratory spectral cost.

The terminal marker removes the initially enabled latch assumption inside this cell. It leaves the semantic frontend port, native joint admission, complete work-word routing and any occupancy sensor OPEN. The Ohmic audit shows why the earlier local certificate cannot be promoted solely by substituting a thermal reverse rate. No microscopic continuum replay, device calibration, hardware validation, fault tolerance or quantum advantage is established. Physical promotion remains zero.

The next useful frontier is a positive coarse-grained or partial-secular generator with an explicit finite-window approximation debit for the same bath. In parallel, compose the terminal port with the declared work word and preserve the original spectators and routing resource ledger. Both are needed before a source-to-storage architecture certificate.

REFERENCES

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